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Influential factors in anaerobic digestion of rice-derived food waste and animal manure: A comprehensive review

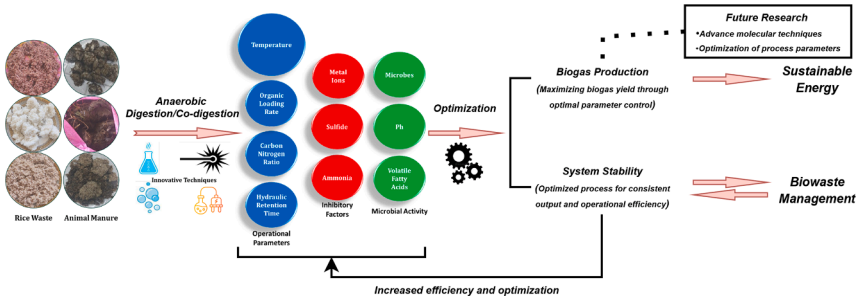
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HIGHLIGHTS

- Anaerobic digestion produces biogas from rice food waste and animal manure.
- Temperature, pH, C/N ratio, microbial activity are crucial for biogas production.
- Micro-aerobic systems and additives boost biogas yield in anaerobic digestion.

GRAPHICAL ABSTRACT



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ABSTRACT

Utilization of organic community wastes towards deriving sustainable renewable energy and adequate disposal of the residual has been an important topic of investigation. Anaerobic digestion and co-digestion of rice-derived food waste and animal manure for sustainable biogas generation is crucial from the view-point of community consumption. This paper presents an extensive review of the important and recent contributions in the related areas. The critical physico-chemical parameters involved in such digestion process are analyzed, including temperature, carbon–nitrogen ratio, microorganisms, pH, substrate characteristics, organic loading rate, hydraulic retention time, volatile fatty acids, ammonia, and light/heavy metal ions. Studies implied that the optimum yield of biogas production could be achieved only when the values of the parameters exist in the specific ranges. Few recent studies highlighted the use of emerging techniques including micro-aerobic system, additives, laser radiation, bio-electrochemical field, among others for efficiency enhancement of the digestion process and optimum yield. The entire study provided a set of important conclusions and future research directives are as well proposed.

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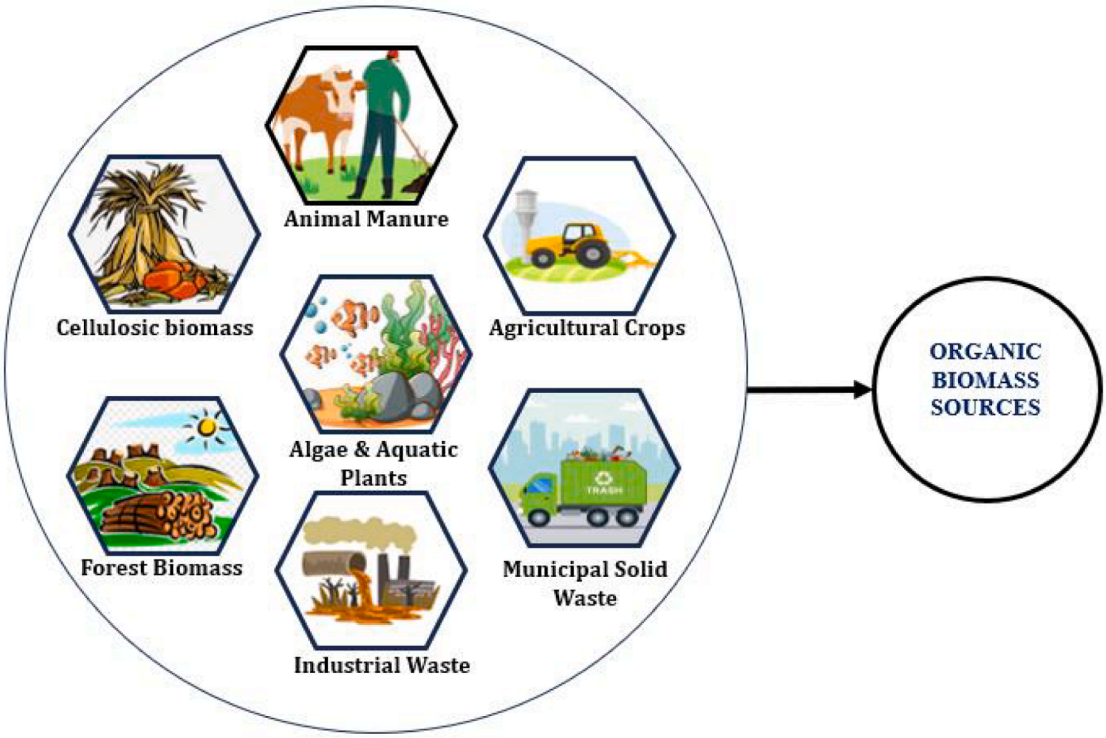


Fig. 1. Various Sources of Organic Biomass.

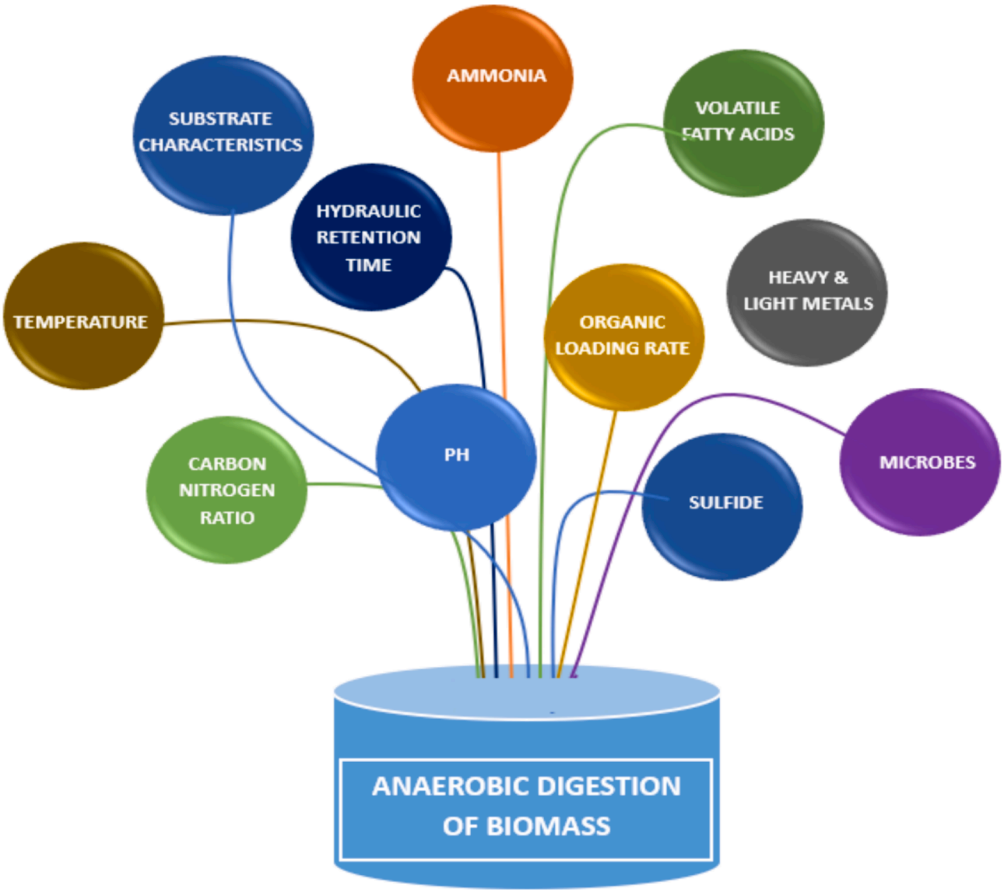


Fig. 2. Various factors affecting Anaerobic Digestion of organic Biomass.

1. Introduction

Rice is a major food in more than 50 % countries worldwide, since it is a source of global nutrition consisting of various proteins, minerals and vitamins, apart from being high in calorie intake due to significant carbohydrate content (Choudhury et al. 2013; Fukagawa and Ziska 2019). More than 90 % of global production of rice originates from Asia, with the majority in China and India (Bandumula 2018; Hashim et al. 2024). Although the white and brown rice are the two foremost types consumed widely for palatability, ease of cooking and durability, other varieties including red, black, and purple rice are sometimes available in the market as well (Veni 2019). In India, the Himalayan region in northern and north-eastern parts, the alluvial and coastal zones in eastern and southern parts contribute to majority of the rice cultivation (Singh et al. 2017). Global rice consumption continues to ascend steadily due to population growth, particularly in Asian and African countries (Samal et al. 2022). The enormous quantity of rice-derived food waste needs effective disposal; similar situation arises for animal manures. To address these issues, the anaerobic co-digestion of food waste and animal manure has emerged as a viable solution. Anaerobic digestion (AD) aided using food waste and animal manure emerges as a sustainable strategy to mitigate such environmental challenges. This process involves pre-treating food waste with animal manure before landfill disposal or composting, yielding biogas as a fuel source and compost as a by-product. Moreover, this approach helps reduce waste volume, inactivates organic and biochemical processes, mitigates landfill-gas emissions and odors, and decreases the required landfill space. By utilizing the environmentally friendly resource of methane and its by-product as fertilizer, AD of food waste presents an alternative to improve the environmental conditions associated with food waste and animal manure while reducing reliance on non-renewable energy sources like wood, coal, and petroleum. There are various sources of organic biomass available (Fig. 1).

AD of organic biomass comprises a complex chain of biological degradation processes controlled by a variety of factors. To properly regulate the controlling factors, a thorough knowledge of the biochemical processes of anaerobic bacteria within the AD system is required (Viotti et al. 2004). Numerous factors have an impact on the performance of microbial consortia in AD, either enhancing or inhibiting the process. These factors include substrate characteristics, organic loading rate, mixing conditions, presence of inhibitors substances, temperature, pH, carbon-to-nitrogen ratio, microbial composition, and retention time, (Fig. 2).

A critical review of the existing literature reveals substantial gaps in the comprehensive understanding of the synergistic effects of various operational parameters on the AD process, particularly in the context of rice-derived food waste. While different studies have addressed the AD of organic wastes, the specific challenges associated with rice food waste, characterized by its unique composition and degradation properties, have been less explored (Viotti et al. 2004). Additionally, the variability in the performance of AD across different climatic and geographical regions, especially in areas with high rice consumption and production, necessitates a more localized optimization of the process parameters. Moreover, there is a noticeable deficiency in studies focusing on the advanced modeling and simulation of the anaerobic digestion process that incorporates the dynamic interactions between microbiological, chemical, and physical factors (Choudhury et al. 2013). Such modeling efforts are crucial for predicting the process performance under varying operational conditions, thereby facilitating the design and optimization of more efficient and robust anaerobic digestion systems (Lucchi, 2023). Lastly, the integration of anaerobic digestion with other waste management and energy production strategies remains underexplored (Bandumula 2018). The potential for coupling anaerobic digestion with technologies such as pyrolysis, gasification, and composting to achieve a zero-waste approach and enhance energy recovery from rice food waste has not been fully realized or documented in the literature. To overcome these gaps, this study investigates the combined influence of these critical factors on specific growth rates, degradation rates, biogas production, and substrate utilization in the context of AD of rice food waste and animal manure. Through a detailed analysis of the interaction between these variables and microbial activity, this study seeks to delineate optimal strategies to enhance the efficiency and effectiveness of the AD process, thereby contributing to the quest for sustainable solutions for food waste management and renewable energy production.

The critical parameters influencing anaerobic and co-digestion efficiency include temperature, pH, carbon–nitrogen ratio, microbial communities, substrate characteristics, organic loading rate, hydraulic retention time, volatile fatty acids, ammonia, and light/heavy metal ions (Fig. 2). Each of these factors plays a pivotal role in determining the process efficiency and stability, particularly in co-digestion scenarios where food waste and animal manure are combined. Numerous interrelated factors have an impact on the biogas yield and understanding the impact of these parameters is essential for optimizing biogas production and enhancing the sustainability of the digestion systems. The rationale for reviewing these factors lies in the need to establish a deeper understanding of their effects on the microbial processes involved in anaerobic digestion or co-digestion, which can lead to more efficient and scalable biogas generation technologies (Mhlanga et al. 2023; He et al. 2024).

2. Methodology

This literature review adopts a systematic and rigorous methodological framework to explore the state-of-the-art in anaerobic digestion processes, specifically focusing on the treatment of rice-derived food waste and animal manure. Extensive literature survey was conducted across standard scientific databases including Scopus, Google Scholar, Research Gate, Science Direct, Springer, etc. Reference lists of identified articles were also examined to capture additional pertinent studies. Inclusion criteria to filter relevant studies included the focus on rice food waste or similar organic waste substrates in AD processes, and publications describing optimizations strategies, technological advances, and novel microbial treatments. Exclusion criteria included studies focusing solely on aerobic treatment methods.

Information from selected articles have been summarized into various categories related to anaerobic digestion, to review different influencing factor including temperature, carbon nitrogen ratio, microbes, pH, substrate characteristics, hydraulic retention time, and the specific chemical compounds like volatile fatty acids, ammonia, sulfide, light and heavy metal ions.

This structured approach facilitates a comparative analysis across different studies, enabling the identification of trends, discrepancies, and areas lacking sufficient research.

3. Influencing parameters in anaerobic digestion

The critical parameters influencing the anaerobic digestion have been briefly described in this section.

3.1. Temperature

Temperature selection and control are fundamental steps in the AD process as they directly influence microbial activity. Hydrolysis and methanogenesis rates in anaerobic decomposition are controlled by temperature. Furthermore, it has an influence not only on the metabolic processes of microbial communities but also on settling properties and gas transfer rates of biosolids. There are three distinct temperature ranges commonly used in AD: psychrophilic (0–25 °C), mesophilic (20 to 45 °C), and thermophilic (above 50 °C) with widespread use of mesophilic and thermophilic conditions (Khalid et al. 2011; Mathew et al. 2014; Rosinska et al. 2016). The microbial communities' morphologies

vary significantly at the two optimum temperatures. According to Nie et al. (2021), temperature has a notable impact on the hydrolysis rate constant as it alters the activities of extracellular enzymes and shifting from psychrophilic to mesophilic conditions results in an increase in both the quantity and variety of volatile fatty acids produced. There exists a variation of optimum temperature for the proliferation of different microorganisms involved in AD contribute to the effect of temperature on the qualitative and quantitative aspects of the desired products, such as H_2, VFAs, and CH_4. Unless necessary microbial communities have proliferated adequately, a temperature shift from mesophilic to thermophilic might result in a significant decrease in biogas generation (Chae et al. 2008). An innovative low-temperature, closed-loop AD has been recently conducted by Bele et al. (2024) who performed tests at 19 °C − 21 °C to get favorable results.

3.2. Carbon-to-nitrogen ratio

Anaerobic digestion process is critically influenced by carbon-to-nitrogen (C/N) ratio of organic material, impacting its efficiency (Mathew et al. 2014; Shahbaz et al. 2020). The optimum C/N ratio for effective anaerobic degradation ranges from 20 to 35 (Lee et al. 2009; Mathew et al. 2014; Wang et al. 2012; Wang et al. 2013; Wang et al. 2014; Hagos et al. 2017; Xu et al. 2018). However, feedstocks often deviate from this range needing co-digestion to optimize the ratio (Xu et al. 2018; Shahbaz et al. 2020; Ghaleb et al. 2021). Co-digestion provides a better balance of substrate nutrients compared to mono-digestion, leading to improved C/N ratios, enhanced buffer capacity, dilution of toxic substances, increased biodegradation, moisture adjustment, and supplementation of trace elements. These changes in nutritional content have an impact on the structure of microbial community and synergistic effects of fermentative bacteria and methano archaea. Biogas production improves when Positive interactions between co-substrates and microbial populations occurs. When rice straw, with a high C/N ratio, is mixed with manures possessing lower C/N ratios, the resulting mixtures exhibit improved C/N ratios compared to single substrates (Wang et al. 2013; Zaher et al. 2009). Furthermore, increased methanation in these mixes can be related to microorganism synergy (Agdag et al. 2007). A lower C/N ratio in food waste promotes microbial accumulation and nitrogen breakdown during deliberate substrate supply disruptions, with microbes showing a preference for consuming leftover nitrogen-bearing protein over readily biodegradable carbohydrates, resulting in higher biomass production (Salangsang et al. 2022). A recent study found that the optimum range of C/N ratio of AD of several animal manures has been 2.3–12.96 (Kadam et al. 2024).

3.3. Microbes

The collaborative activities of various microbial communities are significant to produce biogas and maintain stable system. The AD process begins with hydrolytic microbes that break down complex organic molecules into simpler ones. Acidogens breakdown these mono and oligomers into volatile fatty acids, they undergo subsequent conversion into acetic acid, CO_2, and H_2 by acetogens. The final phase, methanogenesis, involves acetoclastic and hydrogenotrophic archaea which convert acetic acid or CO_2/H_2 into methane (Heeg et al. 2014). Recent advancements have significantly expanded our understanding of the microbial ecology in AD. New high-throughput sequencing technologies, such as Next-Generation Sequencing (NGS) and metagenomics, have provided deeper insights into the microbial dynamics and community structures involved in AD processes (Carabeo-Pérez et al. 2019). These techniques enable a more detailed and comprehensive analysis of microbial species and their metabolic pathways, offering the potential to enhance biogas yields by better controlling microbial consortia. Polymerase chain reaction based ribotyping (culture-independent methods) have been successful in studying the microbial populations implicated in AD (Dzofou Ngoumelah et al., 2024). Methanogens, which are usually unculturable microorganisms, have been extensively studied using these techniques (Wirth et al. 2012; Chojnacka et al. 2015). By employing culture-independent methods, researchers have identified constituents of more than 20 bacterial phyla present in anaerobic waste and wastewater sludge, with proteobacteria, chloroflexi, firmicutes, and bacteroidetes being the most prominent ones, whereas methanogens, especially thermophilic strains, dominate the archaeal community (Chouari et al. 2005; Chojnacka et al. 2015; Goswami et al. 2016). Recent studies have highlighted the importance of syntrophic relationships between acetogens and methanogens in ensuring efficient biogas production. Disruptions in these relationships due to changes in pH, temperature, or substrate characteristics can lead to suboptimal performance or even process failure (Yuanyuan et al., 2012). Key methanogens include methanosarcina barkeri, methanosarcina frisius, and methanobacterium formicicum, along with thermophilic species such as crenarchaea or thermoplasma sp., which have been identified in a variety of AD environments (Godon et al. 1997). Further developments in bioaugmentation, where specific microbial strains are introduced into AD systems to improve performance, have shown promise in enhancing stability and resilience, especially in systems subjected to high organic loading or environmental stressors (Kaur et al. 2024). For example, microbial consortia enriched with thermophilic methanogens have demonstrated increased methane yields and process robustness in high-temperature digesters (Ryue et al. 2020; Singh et al. 2023). In co-digestion process, firmicutes and chloroflexi have been found to exhibit extensive organic compound degradation capabilities. The role of firmicutes in protein and carbohydrate degradation has been increasingly recognized, with their prevalence correlating strongly with higher digestion efficiency and enhanced microbial diversity (Scott et al. 2013). The bacterial population in AD systems consists of 2331 operational taxonomic units grouped into 24 phyla and 263 genera, with firmicutes being the dominant phylum, constituting approximately 89.2 %. Reactors with a higher proportion of firmicutes consistently achieve better digestion performance. Within the archaeal community, a total of 6783 OTUs have been classified into three phyla, predominantly euryarchaeota, including methanogens such as methanosarcina, methanosaeta, methanoculleus, and methanobrevibacter (Xu et al. 2018; Xue et al. 2019). New findings suggest that the stability of these archaeal communities is key to maintaining optimal methane production, particularly under varying environmental conditions (Wang et al. 2024). These developments underscore the complex and dynamic nature of microbial ecosystems in AD, emphasizing the need for continued research and innovation to further improve process efficiency and stability.

3.4. pH

The performance and stability of an anaerobic digester, as well as the modification of reaction rates for individual sub-processes, largely depends on the pH level of the content (Lohani et al. 2018). An efficient-functioning AD process proficiently converts the products of each metabolic cycle into the next without considerable buildup of intermediate metabolites that might cause a pH decrease, such as different fatty acids. The inclusion of certain alkalinity sources increases the pH-buffering capacity and decreases the inhibition of the methanogenic process caused by food waste (FW). This shows that alkalinity sources with required amounts of alkalinity and microelements help to stabilize processes and increase methane generation (Chen et al. 2015). The pH value influences the activity of hydrolytic enzymes, which is important in anaerobic biodegradation (Mathew et al. 2014). Optimal initial pH conditions (neutral to basic) stimulate alkaline protease-catalyzed hydrolysis, which reduces sludge solids and solubilizing organic matter (Pang et al. 2023). The formation of CH_4 is hindered by an Acidic pH. Increasing pH from 5.0 to 7.5 improves CH_4 formation while CO_2 production remains stable, while pH 7.0 and 7.5 enable stable and reproducible digestion in AD of water buffalo manure (Carotenuto et al.

2020). An initial pH of 7.5 resulted in the maximum methane production and volatile solids degradation rate in co-digestion of kitchen waste and cow manure (Zhai et al. 2015). The initial pH and fermentation time of the AD process had distinct and substantial impacts on the production of volatile fatty acids, gibberellic acid, and indole-3-acetic acid (Castro et al. 2022).The breakdown of solid organic matter into soluble compounds and development of microorganisms are influenced by the prevailing pH range(Kim et al. 2003; Mahmoud et al. 2004). The most effective range of pH for methanogenesis, the final phase of AD is found to be 6.5–8.2. The study also found that changing the initial pH had a significant influence on pH, alkalinity, volatile fatty acids, and NH_4^+ N^- variations. Altering the pH at the start of co-digestion effectively increased methane generation. Among the treatments, pH 7.5 produced the best results for co-digestion of kitchen waste with cow dung. Al-Zoubi et al. (2024) observed that optimized biogas production at a pH range of 6–8.

3.5. Substrate characteristics

The effectiveness of AD process on the growth rate of anaerobic bacteria, yielding of biogas and rate of degradability is determined by the nature of organic matter content of the feedstock. The issue of odour pollution from municipal solid waste treatment plants has grown in popularity among the general population. Animal manure, organic fractions of municipal solid waste, sewage sludge, agricultural waste, rural and abattoir effluents, among other waste sources with significant levels of biological contamination, can all be used as feedstock (Gelegenis et al. 2007; Karagiannidis et al. 2009). According to Bong et al. (2018), the variations in the physical and biochemical properties of food waste can impact the performance of the applied treatment for process improvement, such as nutritional balancing, mechanical Mata treatment, thermal treatment, and two-stage design. Research done in the United States looked at the possible benefits of employing cattle carcasses in small-scale reactors, which revealed increased methane (CH_4) and carbon dioxide (CO_2) production when compared to solid waste (Yuan et al. 2012). It was discovered that enzymatic pre-treatment of cattle manure increased biogas and methane output (Yen et al. 2007). Carbon dioxide evolution was used to stabilize the anaerobic digestate (Trzcinski et al. 2011). Waste incineration has been offered as a viable means of reducing waste bulk and volume (Lombardi et al. 2012). Different microbial consortiums consume the elements contained in the feedstock fed to the digester. Furthermore, for the synthesis of amino acids, proteins, nucleic acids, and the neutralization of volatile fatty acids (VFAs) nitrogen is required in the feedstock. However, too much nitrogen in the feedstock might create harmful consequences due to ammonia production. As a result, to provide adequate nutrients, a proper nitrogen content is required while avoiding ammonia toxicity (Hagos et al. 2017; Vindis et al. 2019). The viscosity of the substrate significantly impacts AD performance and microbial communities. Manipulating sludge viscosity through stirring rate adjustments can serve as a control strategy to boost biogas production in anaerobic digesters (Cao et al. 2023). According to Ulukardesler (2023), the biodegradability of substrate plays a vital role on AD process and biogas yield. In an anaerobic digester, the presence of key components such as lipids, proteins, carbohydrates (such as cellulose and hemicelluloses), and lignin, determined the Solid waste degradability and biogas generation potential (Hartmann et al. 2000). Lipids are particularly significant in AD because they produce more methane than other organic compounds. However, Solid waste such as kitchen trash containing high levels of lipids decrease hydrolysis rates due to probable synergistic effects and lipid adsorption. Lignocellulosic waste, such as agricultural waste, garden waste, and paper residue, is abundant but due to presence of lignin it is resistant to anaerobic digestion. Hydrolysis, the rate limiting phase in the whole process poses a significant challenge in dealing with lignocellulosic waste. The composition of waste determines the C/N ratio, with a high C/N ratio suggesting a lack of nitrogen for bacterial growth, resulting in low gas production rates and solids degradability. Contrarily, A high C/N ratio results in ammonia accumulation, which is poisonous to bacteria (Hartmann et al. 2000). Particle size plays an important role during hydrolysis, the smaller the size of particles more is the surface area for enzyme action (Hartmann et al. 2000). Evidently, in AD of food waste, increasing the average particle size reduces the maximum substrate utilization rate coefficient, contrarily, lowering the particle size to 2 mm improved methane generation and fiber degradation when compared to untreated fibers (Hartmann et al. 2000). As per Chen et al. (2008), the levels of biomass, toxicant, exposure duration, cell age, feeding behaviors, acclimatization, and temperature significantly affect the concentration of hazardous organic compounds. AD is hindered by significant organic molecules like chlorophenols, halogenated aliphatic compounds, nitrogen-substituted aromatic compounds, long-chain fatty acids, and lignin/lignin-related chemicals. Due to their unique physicochemical properties, organic wastes like drinking sludge obtained from the paper recycling industry face restrictions in their application within AD processes (Bareha et al. 2022).

3.6. Hydraulic retention time

Hydraulic retention time (HRT) is defined as the retention time of waste material in the digester, and theoretically it is equal to the ratio of the digester-volume to the daily intake flow rate (Srisowmeya et al.. 2020). HRT implied the available duration for the microbial growth and for bioconversion of the wastes to biogas, which primarily depends upon the type of substrate, especially its concentration and microbial concentration, and the operation condition, including temperature, pH and loading rate (Haryanto et al. 2018). The influence of decreasing HRT in an anaerobic membrane bioreactor process has been studied by many researchers. Wang et al. (2018) conducted performance study on a pilot-scale anaerobic membrane bioreactor under short HRT for municipal wastewater. Similar study on cattle slaughterhouse wastewater was conducted by Musa and Idrus (2019). The affinity of methanogens in treating low strength acetate wastewater on decreasing HRT was performed by Hou et al. (2020). During reduced HRT processes, significant shift of bacterial community was observed, while proliferation of acid-producing bacteria resulted in accrual of volatile fatty acid. HRT was also found to influence biogas yield, subtraction efficiency, and transitional metabolism (Pramanik et al. 2020). For the anaerobic process if the HRT is reduced, the system may collapse owing to the washout of these vital microbes required (Zaher et al. 2009). Lowering the HRT can result in smaller digesters and reduced construction costs. A shorter HRT, alternatively, results in a faster biogas generation rate but less effective organic matter decomposition, such as volatile solids or COD. This lack of process stability relates to a decline in process efficiency. Subject to waste type, operation temperature, process stages, and digester structure, HRT values for solid waste anaerobic digesters have been reported ranging from 3 to 55 days (Hartmann et al. 2000). Dry AD HRTs typically vary from 14 to 30 days, however wet AD HRTs can be as short as 3 days. In contrast, a digester digesting solid waste from poultry slaughterhouses may require a longer retention period of 50 to 100 days. Shorter retention intervals (13 to 25 days) also hinder the process, as evidenced by the synthesis of long-chain fatty acids and lower methane output. Continuous laboratory studies on Performance enhancement of leaf vegetable waste in two-stage anaerobic systems with short hydraulic retention time and high recirculation rates resulted in methanogenic reactor instability. However, increasing the recirculation rate and the retention duration further resulted in improved overall performance (Zuo et al. 2015).Thermal and alkaline-thermal pretreatments were studied for their capacity to increase AD of sludge at various hydraulic retention time. The enhancing benefits of both pretreatment procedures on organic matter removal increased when the HRT was lowered. However, the enhancing effects on methane output declined when HRT was reduced (Liu et al. 2020).

3.7. Organic loading rate

To reduce biogas generation concerns, different approaches have been tried, like reducing the Organic Loading Rate (OLR) and employing alkali chemicals such as NaOH and CaO (Kleybocker et al. 2012). Organic matter removal rate represents the quantity of organic matter expressed as volatile solids that a specified volume of anaerobic digester is required to handle in each time. The value of OLR is significantly connected to the HRT. The OLR is computed by dividing the substrate's organic matter content by the reactor's liquid volume and the substrate feeding rate. A shorter HRT results in a larger OLR when the feedstock's organic matter concentration remains constant. However, even at the same HRT, fluctuations in the organic matter concentration of the feeding substrate can cause variations in the OLR. The rapid intermediate product formation by hydrolysis and acidogenic bacteria is one possible risk of high OLR. Methanogenic bacteria, due to their slower multiplication rate, may face challenges to ingest the fatty acids to the same extent. Subsequently, the accumulation of fatty acids leads to decrease in pH and impedes the functionality of methanogenic bacteria, eventually leading to system failure. The two-stage process using organic fraction of municipal solid waste and food processing waste outperformed the single-stage process, allowing for greater organic loading rates (OLRs) and shorter hydraulic retention times (HRTs) which increased stability and performance(Aslanzadeh et al. 2014). Temperature and OLR had a substantial impact on biogas generation, intermediate metabolism, and degradation performance (Liu et al. 2017). mesophilic and thermophilic reactors functioned reasonably well at low organic loading rates (OLRs). However, at higher OLRs, the thermophilic condition had a considerable negative impact on AD performance (Bi et al. 2019). Organic fraction of Municipal Solid Waste (OFMSW) demonstrated stable operation at OLRs ranging from 0.75 to 9.00 g VS/(Ld) under both mesophilic and thermophilic environments (Jiang et al. 2020). It was discovered that methane output did not increase linearly with increasing OLR, highlighting the significance of OLR in both laboratory tests and biogas plant design (Duan et al. 2019). Furthermore, successful bioaugmentation was dependent on OLR, with larger levels assuring positive results (Christou et al. 2021). The biogas yield of cow manure exhibited increase with ascending OLR, optimized at methane yield of 125.98 LCH_4/KgVS for a substrate concentration of 40 gVS/L (Alkhrissat 2024).

3.8. Chemical compounds

3.8.1. Volatile fatty acids

As per Rubiaet al. (2019), AD comprises four stages: hydrolysis, acidogenesis, acetogenesis, and methanogenesis. The initial three phases generate volatile fatty acid (VFA) such as acetate, propionate, *iso*-butyrate, n-butyrate, *iso*-valerate, and n-valerate. Among these VFAs, acetate exhibits the most efficient acid, subsequently butyrate, valerate, and propionate (Elefsiniotis et al. 2007). Acetic, propionic, and butyric acids are often identified in acid-phase reactors handling dairy effluent, and their concentrations fluctuate according to operation conditions (Fang and Yu, 2000). These three VFAs are detected in greater quantities than any other VFA.A considerable number of VFAs are formed during the hydrolysis/acidogenesis cycle, upsetting the pH equilibrium. The VFAs accumulation leads to pH decrease, which impairs the activity and equilibrium of acetogens and methanogens, eventually leading to system collapse (Kim et al. 2003; Dogan et al. 2009; Komemoto et al. 2009; Jiang et al. 2013). The study findings indicated the stability of archaeal communities despite fluctuations in VFA levels (Franke-Whittle et al., 2014), highlighting the significance of using VFA concentrations for process control (Lee et al. 2015). Furthermore, the inhibitory effects of VFAs on cellulolytic activity and cellulose hydrolysis were demonstrated (Siegert and Banks, 2005), along with the potential of FeO to convert VFAs from non-acetic to acetic and enhance CH_4 yield through methanogen activity (Kong et al. 2018). The study also identified variations in heavy metal toxicity on VFA degradation, emphasizing the need for metal inhibition strategies (Lin, 1992). During food waste fermentation, the anaerobic activated sludge exhibited higher VFA production (Wang et al. 2014). The recovery of VAF with dewatering, filtration, and extraction units was found to be impacted by several factor including pH control and bioaugmentation (Zhao et al. 2020). Performance study of anaerobic digestion revealed significant effects of free ammonia on VFA accumulation, wherein their co-accumulation inhibited steady state (Shi et al. 2017). The production and accumulation of VFAs during thermophilic AD of sewage sludge demonstrated the dominance of acetic acid and its association with high methane content (Al-Sulaimi et al., 2022). Moreover, optimal pH conditions for VFAs production from potato peel waste were determined, highlighting the stability and efficiency of mixed-acid fermentation (Lu et al. 2020). Finally, the study showcased the potential of chicken manure as a valuable source of VFAs, with optimization strategies involving heat-shock pretreatment, higher inoculum to substrate ratio, and uncontrolled pH. Acetate emerged as the primary VFA, and increased inoculum content improved bioconversion efficiency (Yin et al. 2021).

3.8.2. Ammonia

According to Chen et al. (2008), the decomposition of nitrogen-containing compounds such as proteins, nitrogenous lipids, nucleic acids, and phospholipids during the AD of solid waste produces ammonia. Ammonia's inhibitory effects are assumed to be caused by a variety of processes, including changes in intracellular pH, higher energy needs to overcome hazardous circumstances, and inhibition of enzyme activities. Ammonium occurs in solution as ammonium ions and free ammonia. Due to its capacity to readily permeate cell membranes and passively diffuse into cells, free ammonia is known for its significant inhibitory impact, creating proton concentration imbalances and/or potassium deficit (Chen et al. 2008). Ammonia is a significant inhibitor in AD processes, impacting performance and microbial populations (Yenigün and Demirel, 2013). High concentrations of ammonia can hinder methane production, particularly affecting aceticlastic methanogens more than hydrogenotrophic methanogens (Angelidaki et al. 1993). The inhibitory effects of ammonia depend on various factors such as substrate, pH, temperature, seed sludge, and reactor configuration. Managing ammonia concentrations is crucial in anaerobic digestion, with high total ammonia nitrogen (TAN) levels inhibiting methanogenesis while lower TAN concentrations are beneficial. Adding nitrate to digester can enhance methane production up to a certain level, but excessive nitrate can decrease methane yield. Therefore, maintaining appropriate TAN and nitrate levels is vital for optimal AD (Sheng et al. 2013; Jo et al. 2022). The review emphasizes the role of ammonia concentration, particularly NH3, in causing inhibition and potential failure in AD systems, with acetoclastic methanogens being highly susceptible (Chen et al. 2016; Jiang et al. 2019; Puig-Castellví et al., 2020). Bioaugmentation with specific methanogens, like Methanoculleus sp. DTU887, has been successful in improving methane production and reducing volatile fatty acids under high ammonia conditions. Genomic analysis provides insights into the functional contributions of microbial communities and the establishment of a syntrophy-supported food web (Yan et al. 2018). Additionally, the presence of carbon nano-tubes in AD systems can have complex effects on ammonia inhibition, affecting potassium transport, intracellular energy levels, and microbial detoxification (Yan et al. 2019). Understanding ammonia's role, inhibition mechanisms, and microbial responses is crucial for optimizing anaerobic digestion. Acclimation strategies, provision of specific trace nutrients, and careful management of ammonia levels are important for mitigating inhibition effects and enhancing AD performance for diverse waste streams. Guo et al. (2023) studied the influence of ammonia on AD with particular reference to energy flow and electron transfer, observing both recoveries and collapse under ascending ammonia in steps.

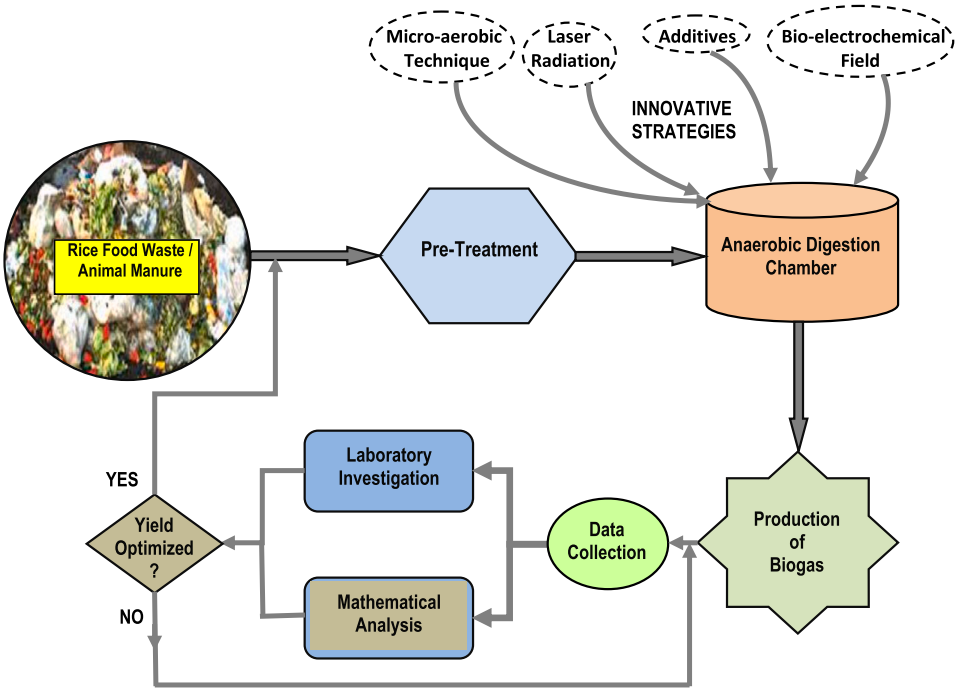


Fig. 3. Illustration of optimization technique in AD.

3.8.3. Sulfide

During anaerobic digestion, the action of sulfate-reducing bacteria (SRB) leads to the reduction of oxidized sulfur compounds and the breakdown of sulfur-containing amino acids, like cysteine, which results in the production of hydrogen sulfide. This reduction process involves two types of SRBs: incomplete oxidizers (acetoclastic SRB) that convert substances like lactate into acetate and CO_2, and complete oxidizers (acetoclastic SRB) that fully transform acetate into CO_2 and HCO_3. Both groups use hydrogen to reduce sulphate. As per Chen et al. (2008), there are two stages of inhibition mediated by sulphate decrease. Primary inhibition is characterized by decreased in methane production because of competition for common organic and inorganic substrates between SRB and methanogenic bacteria. The studies highlighted the inhibitory effects of high concentrations of un-ionized hydrogen sulfide (H_2S) and total sulfide on lactose and acetate utilization, as well as sulfate reduction in AD processes. Additionally, they demonstrated the potential for redirecting carbon flow from sulfate reduction to methanogenesis even in the presence of high sulfate concentrations (Hilton and Oleszkiewicz, 1988). High sulfur load, total ammonia nitrogen (TAN) concentrations, and sulfide levels were found to negatively impact methane production and the performance of anaerobic digesters (Isa et al. 1986; Sürmeli et al. 2019). To address these challenges, various strategies have been explored. These include acclimation to tolerate sulfide and sulfate concentrations, pH adjustment (Yan et al. 2018), in-situ sulfide control technologies (Jung et al. 2022), electrochemical methods (Lin et al. 2016), optimization of micro aeration (Oliveros-Muñoz et al., 2021) and the use of zinc oxide nanowires (Lupitskyy et al. 2018), TiO_2@Ag core–shell nanostructures (Hassan et al. 2023) and inorganic additives (Mutegoa et al. 2021). These approaches aim to improve the efficiency of AD and mitigate the inhibitory effects of H_2S and other contaminants. Furthermore, the studies emphasized the importance of understanding trace metal bioavailability and speciation in AD processes (Thanh et al. 2016). By addressing the inhibitory effects of H_2S and other contaminants, AD can be further developed as a sustainable and efficient waste treatment method. Chen et al. (2023) studied upgrading of AD containing sulfur with conductive materials and improvement of microbial tolerance to sulfides.

3.8.4. Light and heavy metal ions

Although AD has shown promise for waste treatment and energy recovery, the presence of inhibitory substances such as light metal ions, heavy metals, and organics create challenges such as lower methane yield and process instability. Light metal ions such as potassium, magnesium, sodium, and calcium are frequently present in anaerobic digestion, and their proper proportions are critical for microbial development. Excessive amounts, on the other hand, might hinder growth, and high quantities can cause salt toxicity and dehydration of bacterial cells. Heavy metals can promote microbial growth but can accumulate in non-biodegradable and toxic forms, causing enzyme deactivation and replacement of naturally occurring metals in anaerobic settings, with their impact determined by parameters such as metal form, pH, and temperature (Chen et al. 2008). Understanding the intricate interactions and effects of heavy metals in AD is critical for process optimization, since different heavy metals have inhibitory or toxic effects based on a variety of factors (Oleszkiewicz et al. 1990; Mudhoo et al. 2013). Furthermore, heavy, and light metal analyses in plant wastes can assist improve feedstock composition (Bożym et al. 2015). The stability and characteristics of metallic and metal oxide nanoparticles influence their antibacterial actions and potential toxicity, necessitating more study to have a complete knowledge (Yang et al. 2013). To enhance methane fermentation, it is necessary to better understand the suppression of anaerobic processes by toxicants such as heavy metals, light metals, and ammonia (Czatzkowska et al. 2020). The combination of hydrothermal treatment and AD offer promising treatment of sewage sludge, providing insights into heavy metal transformation and distribution (Wang et al. 2021). Moreover, study is required to understand the synergistic impacts of exogenous pollutants such as heavy metal and nano particles, on waste activated sludge digestion and to create mitigation techniques (Luo et al. 2020). For the commercial use of dark fermentative hydrogen generation, inhibitory variables such as inorganic inhibitors (heavy metal ions, light metal ions, ammonia, sulphate, and hydrogen gas) must be addressed (Chen et al. 2021). Furthermore, dissolved organic substance-based amyloid-fibrils-like functional materials can remove heavy metals and recover valuable metals (Yang et al. 2020). The addition of zinc to AD of swine manure enhanced biogas production and the prevalence of antibiotic resistance genes, suggesting the need for

Table 1
Summary of literature related to mesophilic and thermophilic reaction in biogas production.

Substrate	Mesophilic/ Thermophilic	Reference
Cattle manure	The higher ammonia concentration reduced the maximum tolerable temperature. The investigation clearly showed that decreasing the process temperature can be a good option for overcoming ammonia inhibition in anaerobic reactors.	Angelidaki and Ahring, 1994
Cowdung	The temperature has a significant effect on biogas production. Optimum yield can be obtained with both mesophilic (30–40 °C) and thermophilic (50–60 °C) processes. A temperature magnitude of more than 60 °C causes gas production to slow down and eventually stop.	Sambo et al., 1995
Paper pulp liquors	Reported that a high explicit methane yield could be acquired in a thermophilic digester with a long HRT.	Ahn et al. 2002
Fruit and vegetable waste	Thermophilic digesters produced more biogas in comparison with psychrophilic and mesophilic digesters, resulting due to higher degradation efficiency in thermophilic conditions.	Bouallagui et al., 2004
Municipal solid waste	The thermophilic procedure yielded 100 % more methane generation and better volatile solids disposal contrasted with the mesophilic process. However, the thermophilic process is sometimes considered less appealing from the energy perspective since it requires more energy for heating.	Zaher et al. 2007
Solid waste streams	Although it requires a longer retention time, the strength of the mesophilic procedure makes it more famous in the current anaerobic digestion system.	Zaher et al. 2009
Food waste	The solubilization rate and production of biogas both are higher in the case of mesophilic (35 °C and 45 °C) conditions than in thermophilic (55 °C and 65 °C) conditions. The activity of the microorganisms was higher in the mesophilic condition than in the thermophilic conditions.	Komemoto et al. 2009
Food waste	Increased temperatures and gas flow rates both increased the rate of ammonia removal. A rapid ammonia removal caused a rapid drop in pH, which could hinder further ammonia removal. The most constant and stable formation of methane can be accomplished at 32–35 °C.	De la Rubia et al., 2010 Sorathia et al. 2012
Seaweeds: Laminaria digitata	Reactors at a mesophilic temperature (35 °C) produced the highest biogas and methane yields than either thermophilic (45 °C)or psychrophilic (20 °C)digesters.	Vanegas and Bartlett, 2013
Municipal wastewater	Though the gas generation rate increases with increased temperature, it lessens the content of methane.	Dai et al., 2017
Basal anaerobic medium	The mesophilic procedure is steadier compared with the thermophilic procedure because a more extensive assortment of microorganisms favors mesophilic temperatures to thermophilic temperatures.	Yen *et al.* 2020

further research (Zhang et al. 2017). Yu et al. (2023) investigated the influence of metal ions of biogas productivity of diary wastes and found that the productivity of enzymes in AD increased with low dose of metal ions, while high dose bridged electron transfer.

Table 2
Characteristics of substrate.

Parameters	Cooked rice	
	Xue et al. 2019	Xue et al. 2020
TS(%)	42.81 ± 0.72	43.91 ± 0.71
VS(%)	97.63 ± 1.09	42.88 ± 1.09
TN/N(%VS)	1.10 ± 0.63	1.12 ± 0.16
TOC/C(%VS)	46.80 ± 0.19	45.67 ± 0.19
Carbohydrate(%VS)	92.24 ± 0.18	
Protein(%VS)	7.13 ± 1.02	
Lipid(%VS)	0.72 ± 0.99	

NOTE TS, total solid, VS, volatile solid; TOC, total organic carbon; TN, total nitrogen

4. Optimal strategies: Improving ad efficiency

In last few years, several techniques and strategies have been undertaken to improve the efficiency of anaerobic digestion. Calderon et al. (2021) proposed a composite pretreatment method with ferric chloride and nitrite which exhibited multiple advantages including quality enhancement of biogas and residual recovery potential. Li et al. (2021) observed that the efficiency of anaerobic digestion process of cow manure can be significantly improved employing several innovative strategies like mechanical and thermal pretreatments, especially when adequate additives are used, such as, nanomaterials, microbial cultures, carbon-based admixtures, etc. Several emerging techniques including laser radiation, applying bio-electrochemical field, etc., are also expected to yield efficiency enhancement, but requires extensive laboratory investigation for verification and validation prior to commercial application. Obileke et al. (2024) conducted a comprehensive review on the optimization techniques and strategies to improve the biogas yields in anaerobic digestion. They observed that the characteristics of substrate, conversion procedure adopted and the pretreatment method undertaken affect the yield significantly, and incorporating mathematical analysis by artificial neural network technique would be beneficial to suggest the most optimal strategy to be implemented. Li et al. (2024) reviewed various approaches on improvement of anaerobic digestion performance and their mechanism. Optimum results would be achieved if a combined pretreatment with biological, physical chemical procedure is employed. Co-digestion and recirculation technique would be effective, while micro-aerobic technology provides adequate balance between the bacteria and he methanogens. Use of additives, although advantageous, its cost-efficiency and commercial availability would limit the application.

To illustrate the optimization technique for enhancing the biogas production, Fig. 3 has been introduced.

5. General overview and critical analysis

The work presents a comprehensive and systematic review on the anaerobic digestion of organic waste substrates including rice-based food waste and animal manure, covering the primary methodology

Table 3
Proximate composition of rice cultivars (Samyor *et al.* 2016).

Parameters	*Lingkangtakerame (LA; red rice)*	*Umlingame (UA; red rice)*	*Pungpotakerame (PA; white rice)*
Moisture(%)	11.13 ± 0.02^{a}	11.01 ± 0.005^{b}	11.01 ± 0.005^{b}
Ash(%)	0.97 ± 0.01^{b}	1.33 ± 0.57^{c}	0.93 ± 0.05^{a}
Fat(%)	1.80 ± 0.20^{b}	2.60 ± 0.19^{c}	1.77 ± 0.06^{a}
Protein(%)	1.95 ± 0.02^{c}	0.24 ± 0.03^{b}	0.12 ± 0.02^{a}
carbohydrate (mg)	79.01 ± 0.16^{b}	81.37 ± 0.93^{c}	76.44 ± 0.59^{a}
Energyvalue(kJ)	16.34 ± 0.01^{c}	16.29 ± 0.01^{b}	16.23 ± 0.02^{a}
Amylosecontent (%)	6.46 ± 0.025^{b}	11.86 ± 0.04^{c}	5.30 ± 0.04^{a}

Table 4
Optimum Carbon to Nitrogen Ratio for Methane Production (C/N).

Raw Materials	Optimum C/N ratio	References
Swine Waste	15.5/1 to 19/1	Sievers et al. 1978
Dairy Manure, Glucose, and Cellulose	25	Hills, 1979
algal sludge and wastepaper	20–25/1 at 35 °C	Yen et al. 2007
Processed Industrial Food Waste and Sewage Sludge	15, 20 and 30	Siddiqui et al. 2011
Dairy Manure, Chicken Manure, and Wheat Straw	25:1 and 30:1	Wang et al. 2012
Dairy Manure, Chicken Manure, and Rice Straw	25 at 35 °C 30 at 55 °C	Wang et al. 2014
Distillers' grains and swine manure	20–30	Sensai et al. 2014
Chicken litter, yoghurt whey, organic fraction of municipal solid waste, hay grass and wheat straw	26–27.5	Zahan et al. 2018
Cooked rice (CR), waste pork (WP), and plant oil (PO) and cow manure	25 and 30	Xue et al, 2020
paper, cardboard and tissue wastes	25	Shahbaz et al. 2020
Oily-biological sludge;sugarcane bagasse	30 at 35–37 °C	Ghaleb et al. 2021
Poultry slaughterhouse wastes with fruit and vegetable solid wastes	30	Samadi et al. 2022

involved, and the influence of the control parameters, such as temperature, pH, carbon nitrogen ratio, microbes, substrate characteristics, HRT, VFA, ammonia, sulfides and metal ions. Appropriate emphasis on physic-chemical parameters, microbial dynamics in biogas production, and the practical implications including strategic planning for optimal efficiency of biogas yield has also been included in the study.

Based on the existing literature, the influence of mesophilic and thermophilic temperature ranges on AD have been summarized in Table 1 below.

Substrate's characteristics significantly influence the production of biogas and its composition. The summarized standard values of different parameters are presented in Tables 2 and 3.

Numerous research has agreed that the optimum C/N ratio for AD is 20 – 30, deviations from which can disturb the AD process. The best C/N ratios are obtained by combining materials with high and low C/N ratios, like organic solid waste and animal manure, as illustrated in Table 4. Similarly, controlling the microbial community participating in anaerobic degradation is instrumental in regulating the overall process. Bacteria are responsible for breaking down complex compounds into intermediate metabolites, which are then utilized by methanogens. The phenomena have been portrayed in Fig. 4 and Table 5.

6. Novelty

The in-depth review assists in understanding the critical factors involved, which includes temperature, pH, carbon–nitrogen ratio, OLR, HRT, and others, in the AD and co-digestion processes of rice-based food wastes and animal manures. The study emphasizes the importance of microbial community structure in determining the overall biogas yield and AD system stability. By systematically analyzing the interrelationship between these factors, this review provides a clear roadmap for optimizing biogas production systems, particularly in rural and decentralized contexts, contributing to the advancement of sustainable waste management practices and renewable energy production technologies.

7. Conclusions

The study reveals that anaerobic digestion and co-digestion are effective techniques in bio-gas production, in case they are adequately carried out under controlled condition. AD processes are influenced by a range of critical physico-chemical parameters, whose optimizing and understanding the interrelationships between these parameters are essential for enhancing biogas production, reducing operational issues, and stability of AD systems. Two temperature ranges, namely mesophilic (20 °C – 40 °C) and thermophilic (>50 °C) have been identified as providing optimal conditions. The optimal range of pH for AD is 5.5—8.5. The substrate characteristics significantly affects the production of biogas; rice possesses considerably higher carbohydrate

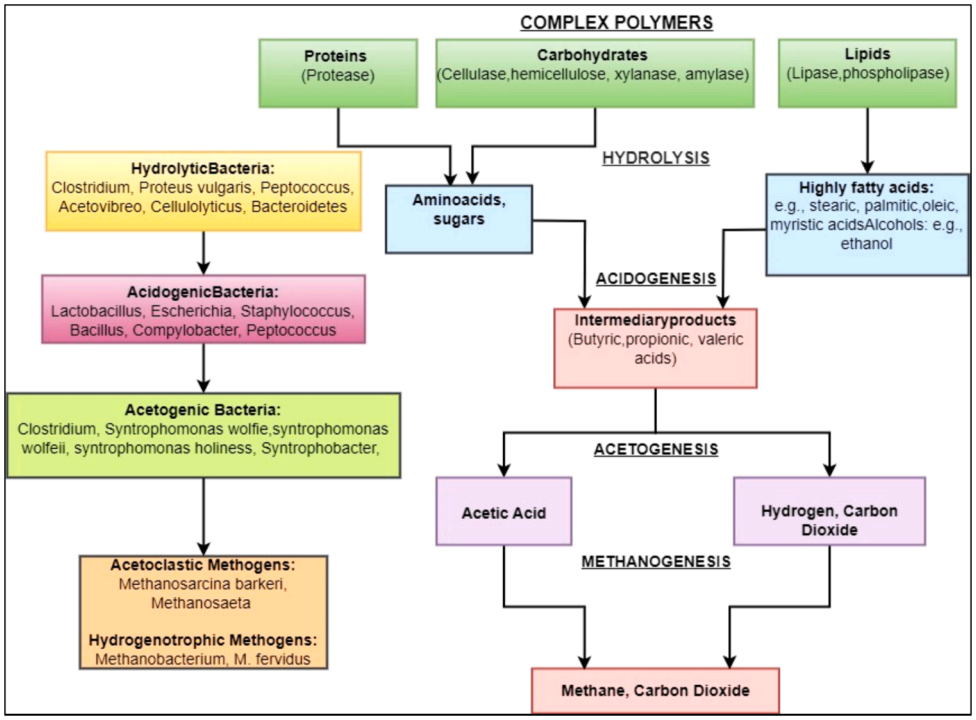


Fig. 4. Carbon flow inside the anaerobic digester and bacteria involved in different stages of anaerobic digestion.

Table 5
Types of bacteria involved in each step of AD of MSW.

Degradation process	Bacterial Group	Type of conversion	Type of Bacteria (Dominant Bacterial Phylum: Proteobacteria, Chloroflexi, Firmicutes, and Bacteroides)
Hydrolysis	Hydrolytic bacteria	Protein to soluble peptides and amino acids	Clostridium, proteus vulgaris, peptococcus Bacteroides, bacillus, vibrio. Clostridium, acetovibrio, cellulitis, staphylococcus, Bacteroides, Clostridium, micrococcus, staphylococcus
Fermentation	Acidogenic bacteria	The carbohydrate to soluble sugar, Lipid to higher fatty acids or alcohol or glycerol	Lactobacillus, Escherichia, Staphylococcus, bacillus, pseudomonas, desulfovibio, selenomonas, sarcina, veillonela, streptococcus, desulfobactor, desulforomonas, clostridium, Eubacterium, limosum, streptococcus
Acetogenesis	Acetogenic bacteria	Amino acids to fatty acids, acetate, and NH3 Sugar to intermediary fermentation products	Clostridium, syntrophomonas wolfie, syntrophomonas wolfeii, syntrophomonas holiness
Methanogenesis	**Archaealgroup** Acetoclastic Methanogens and Hydrogenotrophic Methanogens	Hydrogen and carbon dioxide to methane, Acetate to methane and carbon dioxide	**Archaeal species** Methanobacterium, methanobrevibacter, methanoculleus, methanoplanus, methanospirrilum, methanosaeta, methanosarcina, methanothermococcus

content in comparison to its nitrogen content, which is opposite in case of animal manure. Food waste derived from white rice has higher yield compared to pigmented ones. The optimum carbon–nitrogen ratio for AD is between 20 and 30, since lower value reduces microbial activity while higher value results in less gas generation. Controlling the microbial community participating in AD is instrumental in regulating the overall process, wherein bacteria break down the complex compounds into intermediate metabolites, which are then utilized by methanogens. A longer HRT results in lower loading rate and vice versa. Lowering the HRT and increasing the loading rate could lead to smaller reactor volume, which in turn facilitates cost-effective reactor selection. The optimal strategy for enhancing AD efficiency can be achieved by adopting multiple techniques, including micro-aerobic system, additives, laser radiation, bio-electrochemical field, among others.

8. Future research directives

Future research on the study area should focus on advanced monitoring and control strategies to achieve real-time process optimization, particularly in resource-limited settings. Key areas include further integration of advanced molecular techniques, such as, bioaugmentation and metagenomics, to understand microbial dynamics and metabolic pathways, optimization of process parameters for enhanced biogas production, exploration of alternative feed-stocks, scaling up and efficient system design, development of advanced process models and optimization algorithms, and assessment of the environmental impact of AD residues and by-products (Agostini et al. 2024; Ajayi-Banji et al. 2024). Addressing these aspects will lead to the advancement of sustainable and economically viable waste treatment and renewable energy production technologies.

CRediT authorship contribution statement

Tado Gyadi: Writing – review & editing, Writing – original draft, Visualization, Resources, Project administration, Conceptualization. **Ajay Bharti:** Project administration, Investigation, Formal analysis, Conceptualization. **Sudip Basack:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Conceptualization. **Pradeep Kumar:** Investigation, Data curation, Conceptualization. **Elena Lucchi:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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